

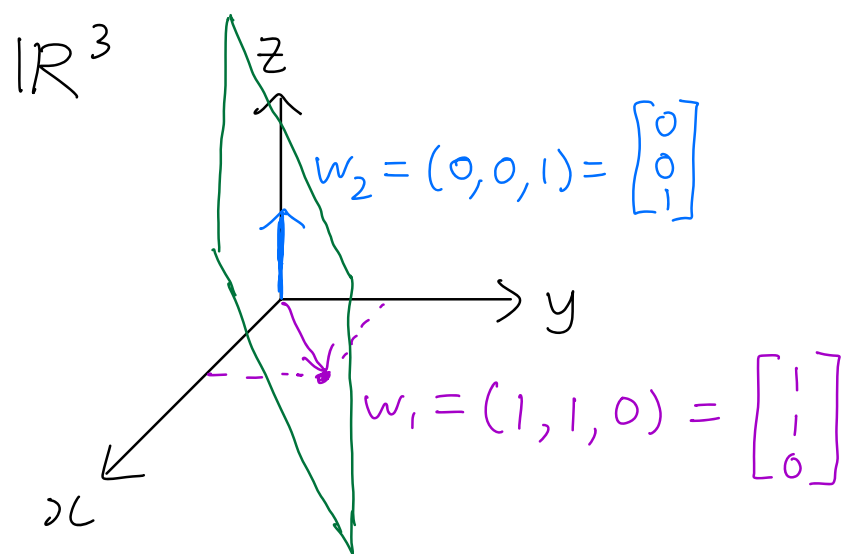
§1.3 Linear Combinations, Span, and Linear Independence

Motivation: We want to construct more subspaces.

Given v.s. V and $w_1, \dots, w_m \in V$, how to make a subspace W that contains $w_1, \dots, w_m$?

(Imagine this as a skeleton of W)

$\Downarrow$



In this case, "visually," guess that

$$W = \mathbb{R}^3 \text{ or}$$

$$W = \left\{ \begin{bmatrix} a \\ -a \\ b \end{bmatrix} : a, b \in \mathbb{R} \right\}$$

$$a \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} =$$

$\Downarrow$

Def 1.30: V v.s.

(1) $v \in V$ is a linear combination ^{l.c.} of $w_1, \dots, w_k \in V$

if $\exists a_1, \dots, a_k \in \mathbb{R}$ s.t. $v = \sum_{i=1}^k \underbrace{a_i}_{\text{coeff of l.c.}} w_i$.

(2) $v \in V$ is a l.c. of S (where $\underbrace{\emptyset}_{\text{empty set}} \neq S \subseteq V$) if

$\exists$ ^{fin} finite subset $\emptyset \neq T = \{w_1, \dots, w_k\} \subseteq S$ s.t. v is

a l.c. of $w_1, \dots, w_k$.

(3) $v \in V$ is a l.c. of $\emptyset \subseteq V$ if $v = 0$.